Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Cross View Dairy

Physical address of dairy:

12530 Warnerville RD	Oakdale	Stanislaus	95361
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1994

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X010-X019-X005-XXXX

B. OPERATORS

Hoekstra, Kurt

Operator name: Hoekstra, Kurt

Telephone no.: (209) 847-5872 Landline, (209) 765-2932 Cellular

12530 Warnerville RD	Oakdale	CA	95361
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Hoekstra, Kurt

Legal owner name: Hoekstra, Kurt

Telephone no.: (209) 847-5872 Landline, (209) 765-2932 Cellular

12530 Warnerville RD	Oakdale	CA	95361
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

Jongsma, Holly

Legal owner name:	Jongsma, Holly	Telephone no.:	(209) 847-2134	
			Landline	Cellular
10007 Country Oak CT	Oakdale		CA	95361
Mailing Address Number and Street	City		State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	350	850	550	145	0
Number under roof	1,575	0	0	0	0	0
Maximum number	1,575	350	850	550	145	0
Average number	1,575	350	850	550	145	0
Avg live weight (lbs)	1,300	1,350	950	700		

Predominant milk cow breed: Other

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 60,202.35 *tons per reporting period*

Total nitrogen from manure: 737,201.71 *lbs per reporting period*

After ammonia losses (30% loss applied): 516,041.20 *lbs per reporting period*

Total phosphorus from manure: 121,753.30 *lbs per reporting period*

Total potassium from manure: 303,696.58 *lbs per reporting period*

Total salt from manure: 822,071.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 93,000,000 *gallons*

Total nitrogen generated: 61,704.57 *lbs*

Total phosphorus generated: 13,716.12 *lbs*

Total potassium generated: 220,088.44 *lbs*

Total salt generated: 1,124,753.95 *lbs*

	93,000,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	93,000,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal OID	Surface water
IW #1	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/03/2023	Solid commercial fertilizer *4-0-0*	13.00 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000
06/29/2023	Solid commercial fertilizer *UN32*	11.70 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	8,519.47	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	8,519.47	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
07/28/2023	Compost	1,540.00 *ton*	Dry-weight	18.4		22,700.00	7,120.00	18,100.00		45.30
09/12/2023	Compost	3,200.00 *ton*	Dry-weight	18.4		22,700.00	7,120.00	18,100.00		45.30

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	175,599.94	55,078.04	140,015.81	3,504,263.04
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	175,599.94	55,078.04	140,015.81	3,504,263.04

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	55	55	1	process wastewater	X010-X019-X005-XXXX
Field 2	40	40	1	process wastewater	X010-X019-X005-XXXX
Field 3	20	20	1	process wastewater	X010-X019-X005-XXXX
Field 4	20	20	1	process wastewater	X010-X019-X005-XXXX
Field 5	30	30	1	process wastewater	X010-X019-X005-XXXX
Field 6	50	50	1	process wastewater	X010-X019-X005-XXXX
Field 7	25	25	1	manure	X010-X019-X005-XXXX
Pivot 3	90	90	2	manure	X010-X015-X061-XXXX
Pivot 4	50	50	2	manure	X010-X015-X061-XXXX
Pivot 5	70	70	2	manure	X010-X015-X061-XXXX
Pivot 6	50	50	2	manure	X010-X015-X061-XXXX
Totals for areas that were used for application	500	500	15		
Totals for areas that were not used for application					
Land application area totals	500	500	15		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

01/01/2023: Pasture

Crop: Pasture Acres planted: 55 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	1,023.00 *ton*	As-is		70.0	10,600.00	890.00	11,700.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	368.00	52.00	378.00	0.00
Total actual harvest content	18.60	394.32	33.11	435.24	1,160.64

Field 2

Field name: Field 2

Field 2

01/01/2023: Pasture

Crop: Pasture Acres planted: 40 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	744.00 *ton*	As-is		70.0	6,000.00	1,060.00	9,060.00		8.75

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	368.00	52.00	378.00	0.00
Total actual harvest content	18.60	223.20	39.43	337.03	976.50

Field 3

Field name: Field 3

01/01/2023: Pasture

Crop: Pasture Acres planted: 20 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	372.00 *ton*	As-is		70.0	14,400.00	1,110.00	10,800.00		9.59

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	368.00	52.00	378.00	0.00
Total actual harvest content	18.60	535.68	41.29	401.76	1,070.24

Field 4

Field name: Field 4

Field 4

01/01/2023: Pasture

Crop: Pasture Acres planted: 20 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	372.00 *ton*	As-is		70.0	9,250.00	1,130.00	9,030.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	368.00	52.00	378.00	0.00
Total actual harvest content	18.60	344.10	42.04	335.92	1,116.00

Field 5

Field name: Field 5

01/01/2023: Pasture

Crop: Pasture Acres planted: 30 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	558.00 *ton*	As-is		70.0	13,900.00	1,300.00	10,700.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	368.00	52.00	378.00	0.00
Total actual harvest content	18.60	517.08	48.36	398.04	1,261.08

Field 6

Field name: Field 6

Field 6

01/01/2023: Pasture

Crop: Pasture Acres planted: 50 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	930.00 *ton*	As-is		70.0	9,470.00	1,090.00	9,430.00		15.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	368.00	52.00	378.00	0.00
Total actual harvest content	18.60	352.28	40.55	350.80	1,674.00

Field 7

Field name: Field 7

11/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 11/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	484.96 *ton*	As-is		73.1	4,840.00	1,020.00	6,740.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	19.40	187.78	39.57	261.49	1,064.51

Pivot 3

Field name: Pivot 3

Pivot 3

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 90 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	998.32 *ton*	As-is		61.6	6,030.00	1,080.00	9,980.00		8.89

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	11.09	133.77	23.96	221.41	757.34

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 90 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	1,835.19 *ton*	As-is		63.2	4,600.00	730.00	3,020.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	20.39	187.60	29.77	123.16	690.36

Pivot 4

Field name: Pivot 4

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	850.06 *ton*	As-is		60.0	6,160.00	1,920.00	19,300.00		9.22

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	17.00	209.45	65.28	656.25	1,254.01

Pivot 4

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	989.20 *ton*	As-is		66.2	4,060.00	840.00	4,270.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	19.78	160.65	33.24	168.96	909.43

Pivot 5

Field name: Pivot 5

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 70 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	1,028.09 *ton*	As-is		57.6	5,980.00	1,740.00	17,800.00		9.17

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	14.69	175.66	51.11	522.86	1,142.08

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	1,560.88 *ton*	As-is		62.0	4,940.00	820.00	3,940.00		4.48

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	22.30	220.31	36.57	175.71	759.21

Pivot 6

Field name: Pivot 6

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	795.39 *ton*	As-is		61.2	5,590.00	1,550.00	14,200.00		9.86

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	15.91	177.85	49.31	451.78	1,217.16

06/06/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/06/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,271.50 *ton*	As-is		57.2	4,840.00	920.00	4,900.00		4.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.43	246.16	46.79	249.21	1,044.87

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 01/01/2023: Pasture

Field name: Field 1

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	143.56	20.25	222.10	1,477.56	1,536,000.00 *gal*
Application event totals		143.56	20.25	222.10	1,477.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.24	0.00	0.00	141.29	5,820,000.00 *gal*
Application event totals		1.24	0.00	0.00	141.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.17	0.00	0.00	134.01	5,520,000.00 *gal*
Application event totals		1.17	0.00	0.00	134.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.19	0.00	0.00	135.46	5,580,000.00 *gal*
Application event totals		1.19	0.00	0.00	135.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.15	0.00	0.00	131.09	5,400,000.00 *gal*
Application event totals		1.15	0.00	0.00	131.09	

Field 1 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.17	0.00	0.00	134.01	5,520,000.00 *gal*
Application event totals		1.17	0.00	0.00	134.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.19	0.00	0.00	135.46	5,580,000.00 *gal*
Application event totals		1.19	0.00	0.00	135.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	138.70	11.33	240.95	1,326.65	1,728,000.00 *gal*
IW #1	Ground water	0.89	0.00	0.00	101.96	4,200,000.00 *gal*
Application event totals		139.59	11.33	240.95	1,428.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.19	0.00	0.00	135.46	5,580,000.00 *gal*
Application event totals		1.19	0.00	0.00	135.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.17	0.00	0.00	134.01	5,520,000.00 *gal*
Application event totals		1.17	0.00	0.00	134.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.20	0.00	0.00	136.92	5,640,000.00 *gal*
Application event totals		1.20	0.00	0.00	136.92	

Field 1 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.19	0.00	0.00	135.46	5,580,000.00 *gal*
Application event totals		1.19	0.00	0.00	135.46	

Field 2 - 01/01/2023: Pasture

Field name: Field 2

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.21	0.00	0.00	138.19	4,140,000.00 *gal*
Application event totals		1.21	0.00	0.00	138.19	

05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.19	0.00	0.00	136.19	4,080,000.00 *gal*
Application event totals		1.19	0.00	0.00	136.19	

05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.16	0.00	0.00	132.18	3,960,000.00 *gal*
Application event totals		1.16	0.00	0.00	132.18	

06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.20	0.00	0.00	137.19	4,110,000.00 *gal*
Application event totals		1.20	0.00	0.00	137.19	

Field 2 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.21	0.00	0.00	138.19	4,140,000.00 *gal*
Application event totals		1.21	0.00	0.00	138.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	143.03	11.68	248.48	1,368.11	1,296,000.00 *gal*
IW #1	Ground water	0.84	0.00	0.00	96.13	2,880,000.00 *gal*
Application event totals		143.87	11.68	248.48	1,464.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.23	0.00	0.00	140.20	4,200,000.00 *gal*
Application event totals		1.23	0.00	0.00	140.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.21	0.00	0.00	138.19	4,140,000.00 *gal*
Application event totals		1.21	0.00	0.00	138.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	137.73	11.25	239.27	1,317.44	1,248,000.00 *gal*
IW #1	Ground water	0.88	0.00	0.00	100.14	3,000,000.00 *gal*
Application event totals		138.61	11.25	239.27	1,417.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.19	0.00	0.00	136.19	4,080,000.00 *gal*
Application event totals		1.19	0.00	0.00	136.19	

Field 2 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.16	0.00	0.00	132.18	3,960,000.00 *gal*
Application event totals		1.16	0.00	0.00	132.18	

Field 3 - 01/01/2023: Pasture

Field name: Field 3

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.23	0.00	0.00	140.20	2,100,000.00 *gal*
Application event totals		1.23	0.00	0.00	140.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	160.10	28.60	305.63	1,659.92	672,000.00 *gal*
IW #1	Ground water	0.91	0.00	0.00	104.15	1,560,000.00 *gal*
Application event totals		161.02	28.60	305.63	1,764.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.19	0.00	0.00	136.19	2,040,000.00 *gal*
Application event totals		1.19	0.00	0.00	136.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.17	0.00	0.00	134.19	2,010,000.00 *gal*
Application event totals		1.17	0.00	0.00	134.19	

Field 3 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.21	0.00	0.00	138.19	2,070,000.00 *gal*
Application event totals		1.21	0.00	0.00	138.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.23	0.00	0.00	140.20	2,100,000.00 *gal*
Application event totals		1.23	0.00	0.00	140.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.19	0.00	0.00	136.19	2,040,000.00 *gal*
Application event totals		1.19	0.00	0.00	136.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	137.73	11.25	239.27	1,317.44	624,000.00 *gal*
IW #1	Ground water	0.95	0.00	0.00	108.15	1,620,000.00 *gal*
Application event totals		138.68	11.25	239.27	1,425.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.17	0.00	0.00	134.19	2,010,000.00 *gal*
Application event totals		1.17	0.00	0.00	134.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.19	0.00	0.00	136.19	2,040,000.00 *gal*
Application event totals		1.19	0.00	0.00	136.19	

Field 3 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.16	0.00	0.00	132.18	1,980,000.00 *gal*
Application event totals		1.16	0.00	0.00	132.18	

Field 4 - 01/01/2023: Pasture

Field name: Field 4

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	209.73	29.59	324.47	2,158.62	816,000.00 *gal*
Application event totals		209.73	29.59	324.47	2,158.62	

04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.23	0.00	0.00	140.20	2,100,000.00 *gal*
Application event totals		1.23	0.00	0.00	140.20	

05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	17.66	8.62	109.95	648.91	2,160,000.00 *gal*
Application event totals		17.66	8.62	109.95	648.91	

05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.23	0.00	0.00	140.20	2,100,000.00 *gal*
Application event totals		1.23	0.00	0.00	140.20	

Field 4 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	17.17	8.38	106.90	630.88	2,100,000.00 *gal*
Application event totals		17.17	8.38	106.90	630.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.19	0.00	0.00	136.19	2,040,000.00 *gal*
Application event totals		1.19	0.00	0.00	136.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	17.42	8.50	108.43	639.89	2,130,000.00 *gal*
Application event totals		17.42	8.50	108.43	639.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.23	0.00	0.00	140.20	2,100,000.00 *gal*
Application event totals		1.23	0.00	0.00	140.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	17.54	8.56	109.19	644.40	2,145,000.00 *gal*
Application event totals		17.54	8.56	109.19	644.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.19	0.00	0.00	136.19	2,040,000.00 *gal*
Application event totals		1.19	0.00	0.00	136.19	

Field 4 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	17.42	8.50	108.43	639.89	2,130,000.00 *gal*
Application event totals		17.42	8.50	108.43	639.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.19	0.00	0.00	136.19	2,040,000.00 *gal*
Application event totals		1.19	0.00	0.00	136.19	

Field 5 - 01/01/2023: Pasture

Field name: Field 5

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	205.62	29.01	318.11	2,116.29	1,200,000.00 *gal*
Application event totals		205.62	29.01	318.11	2,116.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.22	0.00	0.00	138.86	3,120,000.00 *gal*
Application event totals		1.22	0.00	0.00	138.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	17.17	8.38	106.90	630.88	3,150,000.00 *gal*
Application event totals		17.17	8.38	106.90	630.88	

Field 5 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.20	0.00	0.00	137.53	3,090,000.00 *gal*
Application event totals		1.20	0.00	0.00	137.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	17.01	8.30	105.88	624.87	3,120,000.00 *gal*
Application event totals		17.01	8.30	105.88	624.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.24	0.00	0.00	141.53	3,180,000.00 *gal*
Application event totals		1.24	0.00	0.00	141.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	16.85	8.22	104.86	618.87	3,090,000.00 *gal*
Application event totals		16.85	8.22	104.86	618.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.22	0.00	0.00	138.86	3,120,000.00 *gal*
Application event totals		1.22	0.00	0.00	138.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	17.01	8.30	105.88	624.87	3,120,000.00 *gal*
Application event totals		17.01	8.30	105.88	624.87	

Field 5 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.22	0.00	0.00	138.86	3,120,000.00 *gal*
Application event totals		1.22	0.00	0.00	138.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	16.85	8.22	104.86	618.87	3,090,000.00 *gal*
Application event totals		16.85	8.22	104.86	618.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.22	0.00	0.00	138.86	3,120,000.00 *gal*
Application event totals		1.22	0.00	0.00	138.86	

Field 6 - 01/01/2023: Pasture

Field name: Field 6

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	17.17	8.38	106.90	630.88	5,250,000.00 *gal*
Application event totals		17.17	8.38	106.90	630.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	17.12	8.35	106.59	629.08	5,235,000.00 *gal*
Application event totals		17.12	8.35	106.59	629.08	

Field 6 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond B	Process wastewater	41.71	18.93	420.59	1,594.73	5,250,000.00 *gal*
Application event totals		41.71	18.93	420.59	1,594.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	17.03	8.30	105.98	625.47	5,205,000.00 *gal*
Application event totals		17.03	8.30	105.98	625.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond B	Process wastewater	41.35	18.76	416.98	1,581.06	5,205,000.00 *gal*
Application event totals		41.35	18.76	416.98	1,581.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	17.17	8.38	106.90	630.88	5,250,000.00 *gal*
Application event totals		17.17	8.38	106.90	630.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond B	Process wastewater	41.95	19.03	422.99	1,603.84	5,280,000.00 *gal*
Application event totals		41.95	19.03	422.99	1,603.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	17.12	8.35	106.59	629.08	5,235,000.00 *gal*
Application event totals		17.12	8.35	106.59	629.08	

Field 6 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	17.12	8.35	106.59	629.08	5,235,000.00 *gal*
Application event totals		17.12	8.35	106.59	629.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond B	Process wastewater	41.95	19.03	422.99	1,603.84	5,280,000.00 *gal*
Application event totals		41.95	19.03	422.99	1,603.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TW Pond A	Process wastewater	17.08	8.33	106.29	627.28	5,220,000.00 *gal*
Application event totals		17.08	8.33	106.29	627.28	

Field 7 - 11/25/2022: Oats, silage-soft dough

Field name: Field 7

Crop: Oats, silage-soft dough

Plant date: 11/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	241.04	104.04	521.22	3,708.29	370.00 *ton*
Application event totals		241.04	104.04	521.22	3,708.29	

Pivot 3 - 10/30/2022: Oats, silage-soft dough

Field name: Pivot 3

Crop: Oats, silage-soft dough

Plant date: 10/30/2022

Pivot 3 - 10/30/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	165.90	50.34	108.22	1,174.80	1,800.00 *ton*
Application event totals		165.90	50.34	108.22	1,174.80	

10/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.00	0.00	0.00	0.00	9,943,020.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Pivot 3 - 06/01/2023: Corn, silage

Field name: Pivot 3

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	166.32	54.22	304.92	6,497.57	1,080.00 *ton*
Application event totals		166.32	54.22	304.92	6,497.57	

06/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	4.00	0.00	0.00	0.00	
Application event totals		4.00	0.00	0.00	0.00	

06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.90	0.00	0.00	46.55	9,654,930.00 *gal*
Application event totals		0.90	0.00	0.00	46.55	

Pivot 3 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	28.80	0.00	0.00	0.00	
Application event totals		28.80	0.00	0.00	0.00	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.90	0.00	0.00	46.09	9,559,980.00 *gal*
Application event totals		0.90	0.00	0.00	46.09	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.89	0.00	0.00	45.61	9,460,260.00 *gal*
Application event totals		0.89	0.00	0.00	45.61	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.89	0.00	0.00	45.57	9,450,450.00 *gal*
Application event totals		0.89	0.00	0.00	45.57	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.89	0.00	0.00	45.61	9,460,260.00 *gal*
Application event totals		0.89	0.00	0.00	45.61	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.88	0.00	0.00	45.28	9,390,240.00 *gal*
Application event totals		0.88	0.00	0.00	45.28	

Pivot 3 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.87	0.00	0.00	44.70	9,270,360.00 *gal*
Application event totals		0.87	0.00	0.00	44.70	

Pivot 4 - 10/30/2022: Oats, silage-soft dough

Field name: Pivot 4

Crop: Oats, silage-soft dough

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	228.01	98.41	493.05	3,507.84	700.00 *ton*
Application event totals		228.01	98.41	493.05	3,507.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.00	0.00	0.00	0.00	5,532,850.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Pivot 4 - 06/02/2023: Corn, silage

Field name: Pivot 4

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Pivot 4 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	166.32	54.22	304.92	6,497.57	600.00 *ton*
Application event totals		166.32	54.22	304.92	6,497.57	

06/02/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	4.00	0.00	0.00	0.00	
Application event totals		4.00	0.00	0.00	0.00	

06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.89	0.00	0.00	46.05	5,305,550.00 *gal*
Application event totals		0.89	0.00	0.00	46.05	

06/25/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	28.80	0.00	0.00	0.00	
Application event totals		28.80	0.00	0.00	0.00	

06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.89	0.00	0.00	45.57	5,250,400.00 *gal*
Application event totals		0.89	0.00	0.00	45.57	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.88	0.00	0.00	45.36	5,226,100.00 *gal*
Application event totals		0.88	0.00	0.00	45.36	

Pivot 4 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.88	0.00	0.00	45.29	5,218,900.00 *gal*
Application event totals		0.88	0.00	0.00	45.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.88	0.00	0.00	45.24	5,212,250.00 *gal*
Application event totals		0.88	0.00	0.00	45.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.87	0.00	0.00	44.91	5,174,600.00 *gal*
Application event totals		0.87	0.00	0.00	44.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.87	0.00	0.00	44.77	5,158,900.00 *gal*
Application event totals		0.87	0.00	0.00	44.77	

Pivot 5 - 10/31/2022: Oats, silage-soft dough

Field name: Pivot 5

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	228.01	98.41	493.05	3,507.84	980.00 *ton*
Application event totals		228.01	98.41	493.05	3,507.84	

Pivot 5 - 10/31/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.00	0.00	0.00	0.00	7,746,410.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Pivot 5 - 06/05/2023: Corn, silage

Field name: Pivot 5

Crop: Corn, silage

Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	230.03	38.43	81.88	2,720.14	980.00 *ton*
Application event totals		230.03	38.43	81.88	2,720.14	

06/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	4.00	0.00	0.00	0.00	
Application event totals		4.00	0.00	0.00	0.00	

06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.89	0.00	0.00	45.71	7,373,240.00 *gal*
Application event totals		0.89	0.00	0.00	45.71	

06/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	28.80	0.00	0.00	0.00	
Application event totals		28.80	0.00	0.00	0.00	

Pivot 5 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.88	0.00	0.00	45.33	7,311,920.00 *gal*
Application event totals		0.88	0.00	0.00	45.33	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.87	0.00	0.00	44.90	7,243,460.00 *gal*
Application event totals		0.87	0.00	0.00	44.90	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.88	0.00	0.00	45.38	7,320,460.00 *gal*
Application event totals		0.88	0.00	0.00	45.38	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.88	0.00	0.00	45.26	7,300,860.00 *gal*
Application event totals		0.88	0.00	0.00	45.26	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.88	0.00	0.00	45.24	7,297,080.00 *gal*
Application event totals		0.88	0.00	0.00	45.24	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.87	0.00	0.00	45.04	7,265,160.00 *gal*
Application event totals		0.87	0.00	0.00	45.04	

Pivot 6 - 10/31/2022: Oats, silage-soft dough

Pivot 6 - 10/31/2022: Oats, silage-soft dough

Field name: Pivot 6

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	228.01	98.41	493.05	3,507.84	700.00 *ton*
Application event totals		228.01	98.41	493.05	3,507.84	

10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.00	0.00	0.00	0.00	5,527,950.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Pivot 6 - 06/06/2023: Corn, silage

Field name: Pivot 6

Crop: Corn, silage

Plant date: 06/06/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	230.03	38.43	81.88	2,720.14	700.00 *ton*
Application event totals		230.03	38.43	81.88	2,720.14	

06/02/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	4.00	0.00	0.00	0.00	
Application event totals		4.00	0.00	0.00	0.00	

Pivot 6 - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.89	0.00	0.00	46.05	5,305,600.00 *gal*
Application event totals		0.89	0.00	0.00	46.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	28.80	0.00	0.00	0.00	
Application event totals		28.80	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.89	0.00	0.00	45.79	5,275,600.00 *gal*
Application event totals		0.89	0.00	0.00	45.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.89	0.00	0.00	45.77	5,273,900.00 *gal*
Application event totals		0.89	0.00	0.00	45.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.88	0.00	0.00	45.42	5,233,450.00 *gal*
Application event totals		0.88	0.00	0.00	45.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.88	0.00	0.00	45.34	5,223,900.00 *gal*
Application event totals		0.88	0.00	0.00	45.34	

Pivot 6 - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.88	0.00	0.00	45.13	5,199,950.00 *gal*
Application event totals		0.88	0.00	0.00	45.13	

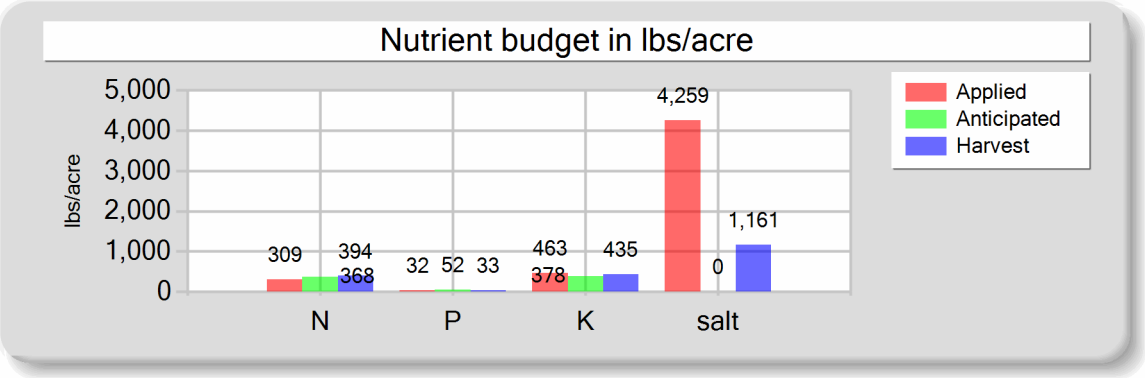
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Surface water	0.87	0.00	0.00	45.03	5,188,900.00 *gal*
Application event totals		0.87	0.00	0.00	45.03	

B. NUTRIENT BUDGET

Field 1 - 01/01/2023: Pasture

Field name: Field 1 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	282.26	31.58	463.05	2,804.21
Fresh water	12.73	0.00	0.00	1,455.13
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	308.99	31.58	463.05	4,259.34
Anticipated crop nutrient removal	368.00	52.00	378.00	0.00
Actual crop nutrient removal	394.32	33.11	435.24	1,160.64
Nutrient balance	-85.33	-1.53	27.81	3,098.70
Applied to removed ratio	0.78	0.95	1.06	3.67

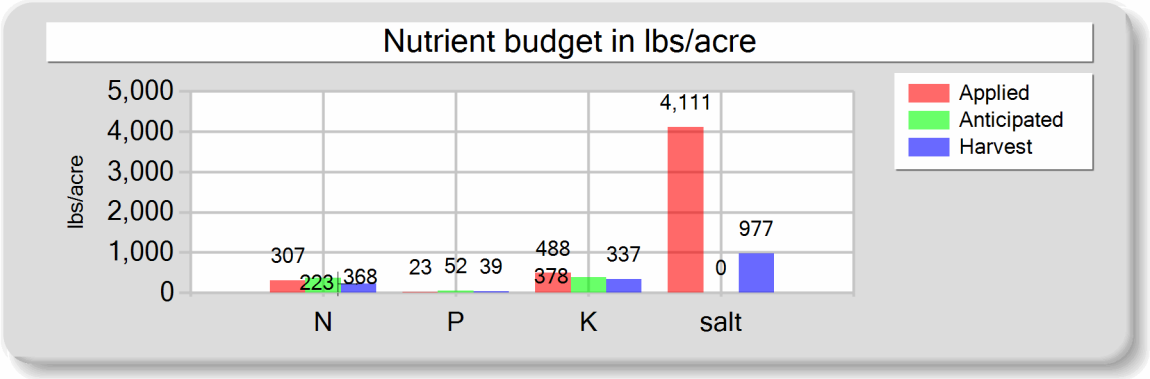
Fresh water applied	
59,940,000.00	*gallons*
2,207.39	*acre-inches*
40.13	*inches/acre*

Process wastewater applied	
3,264,000.00	*gallons*
120.20	*acre-inches*
2.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 01/01/2023: Pasture

Field name: Field 2 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	280.76	22.93	487.75	2,685.55
Fresh water	12.47	0.00	0.00	1,424.99
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	307.23	22.93	487.75	4,110.55
Anticipated crop nutrient removal	368.00	52.00	378.00	0.00
Actual crop nutrient removal	223.20	39.43	337.03	976.50
Nutrient balance	84.03	-16.50	150.72	3,134.05
Applied to removed ratio	1.38	0.58	1.45	4.21

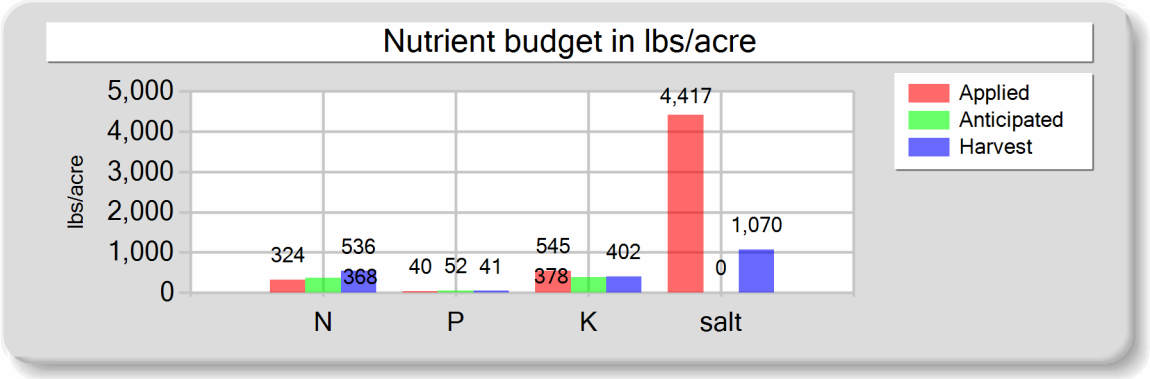
Fresh water applied	
42,690,000.00	*gallons*
1,572.13	*acre-inches*
39.30	*inches/acre*

Process wastewater applied	
2,544,000.00	*gallons*
93.69	*acre-inches*
2.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 01/01/2023: Pasture

Field name: Field 3 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	297.84	39.85	544.90	2,977.36
Fresh water	12.60	0.00	0.00	1,440.01
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	324.44	39.85	544.90	4,417.38
Anticipated crop nutrient removal	368.00	52.00	378.00	0.00
Actual crop nutrient removal	535.68	41.29	401.76	1,070.24
Nutrient balance	-211.24	-1.44	143.14	3,347.13
Applied to removed ratio	0.61	0.97	1.36	4.13

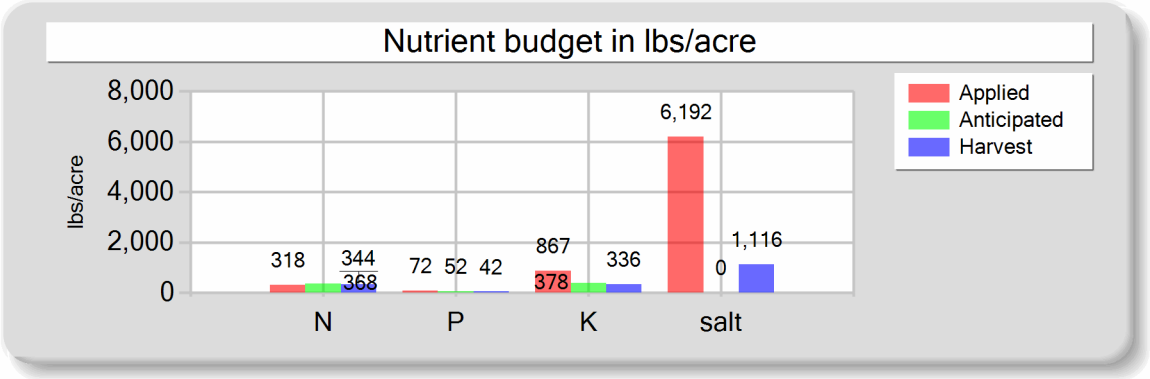
Fresh water applied	
21,570,000.00	*gallons*
794.35	*acre-inches*
39.72	*inches/acre*

Process wastewater applied	
1,296,000.00	*gallons*
47.73	*acre-inches*
2.39	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 01/01/2023: Pasture

Field name: Field 4 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	296.95	72.13	867.37	5,362.60
Fresh water	7.26	0.00	0.00	829.16
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	318.21	72.13	867.37	6,191.76
Anticipated crop nutrient removal	368.00	52.00	378.00	0.00
Actual crop nutrient removal	344.10	42.04	335.92	1,116.00
Nutrient balance	-25.89	30.09	531.45	5,075.76
Applied to removed ratio	0.92	1.72	2.58	5.55

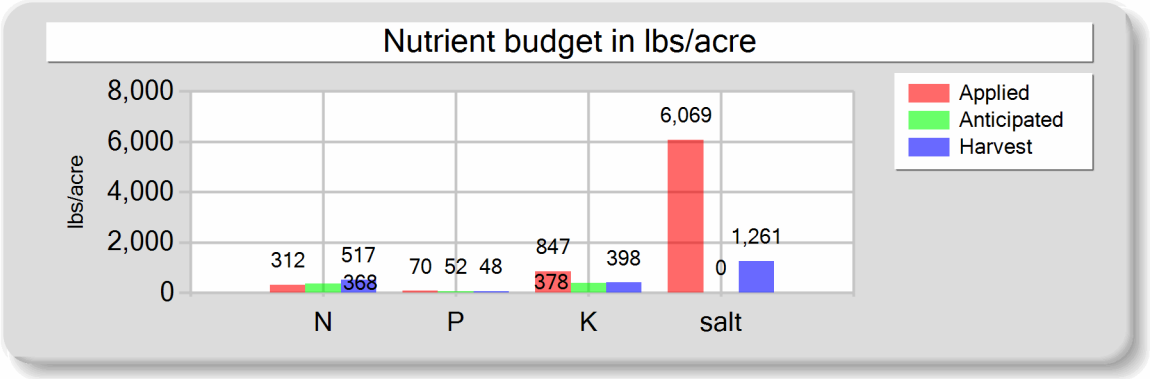
Fresh water applied	
12,420,000.00	*gallons*
457.39	*acre-inches*
22.87	*inches/acre*

Process wastewater applied	
11,481,000.00	*gallons*
422.81	*acre-inches*
21.14	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 01/01/2023: Pasture

Field name: Field 5 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	290.51	70.41	846.50	5,234.65
Fresh water	7.30	0.00	0.00	834.50
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	311.81	70.41	846.50	6,069.15
Anticipated crop nutrient removal	368.00	52.00	378.00	0.00
Actual crop nutrient removal	517.08	48.36	398.04	1,261.08
Nutrient balance	-205.27	22.05	448.46	4,808.07
Applied to removed ratio	0.60	1.46	2.13	4.81

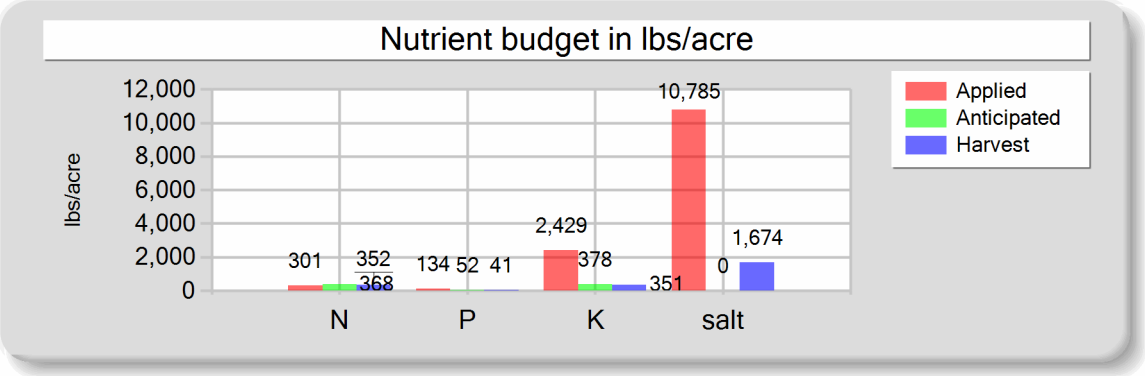
Fresh water applied	
18,750,000.00	*gallons*
690.50	*acre-inches*
23.02	*inches/acre*

Process wastewater applied	
16,770,000.00	*gallons*
617.58	*acre-inches*
20.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 01/01/2023: Pasture

Field name: Field 6 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	286.78	134.21	2,429.41	10,785.23
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	300.78	134.21	2,429.41	10,785.23
Anticipated crop nutrient removal	368.00	52.00	378.00	0.00
Actual crop nutrient removal	352.28	40.55	350.80	1,674.00
Nutrient balance	-51.51	93.66	2,078.61	9,111.23
Applied to removed ratio	0.85	3.31	6.93	6.44

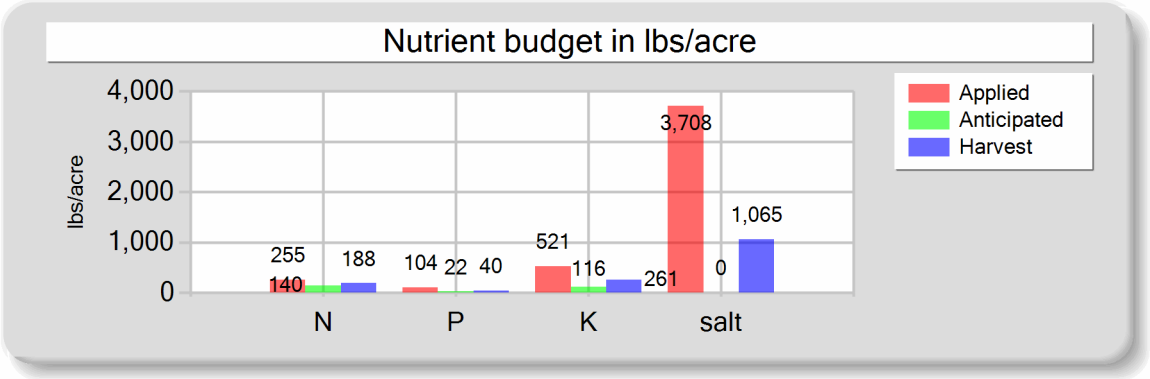
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
57,645,000.00	*gallons*
2,122.87	*acre-inches*
42.46	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 11/25/2022: Oats, silage-soft dough

Field name: Field 7 Crop: Oats, silage-soft dough Plant date: 11/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	241.04	104.04	521.22	3,708.29
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	255.04	104.04	521.22	3,708.29
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	187.78	39.57	261.49	1,064.51
Nutrient balance	67.26	64.47	259.73	2,643.78
Applied to removed ratio	1.36	2.63	1.99	3.48

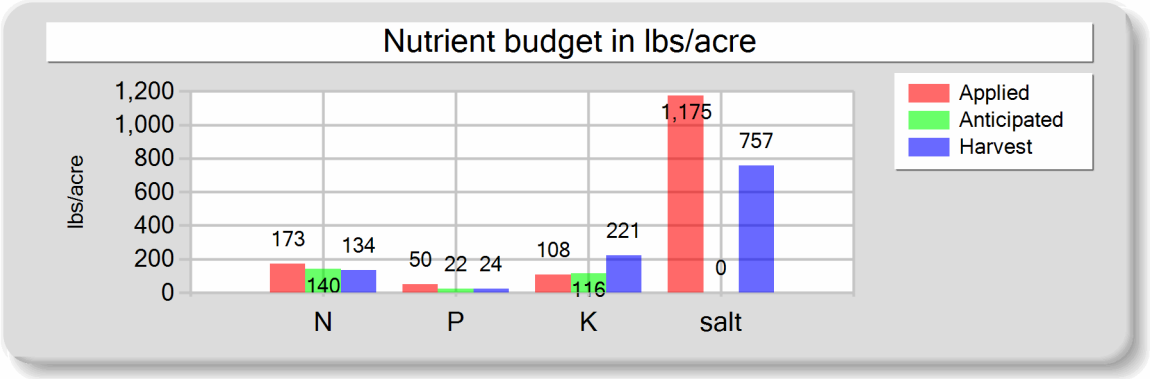
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Pivot 3 - 10/30/2022: Oats, silage-soft dough

Field name: Pivot 3 | Crop: Oats, silage-soft dough | Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	165.90	50.34	108.22	1,174.80
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	172.90	50.34	108.22	1,174.80
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	133.77	23.96	221.41	757.34
Nutrient balance	39.12	26.38	-113.18	417.46
Applied to removed ratio	1.29	2.10	0.49	1.55

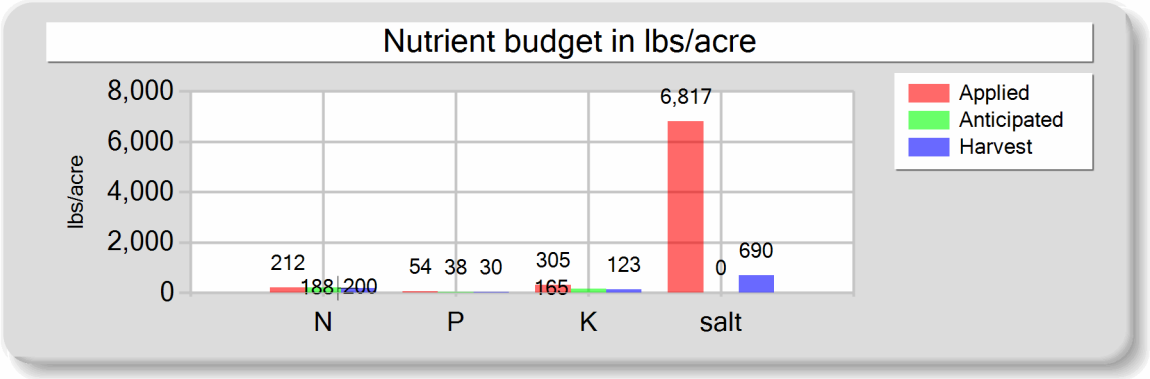
Fresh water applied	
9,943,020.00	*gallons*
366.17	*acre-inches*
4.07	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pivot 3 - 06/01/2023: Corn, silage

Field name: Pivot 3 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.80	0.00	0.00	0.00
Dry manure	166.32	54.22	304.92	6,497.57
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	6.20	0.00	0.00	319.41
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	212.32	54.22	304.92	6,816.98
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	187.60	29.77	123.16	690.36
Nutrient balance	24.73	24.45	181.76	6,126.62
Applied to removed ratio	1.13	1.82	2.48	9.87

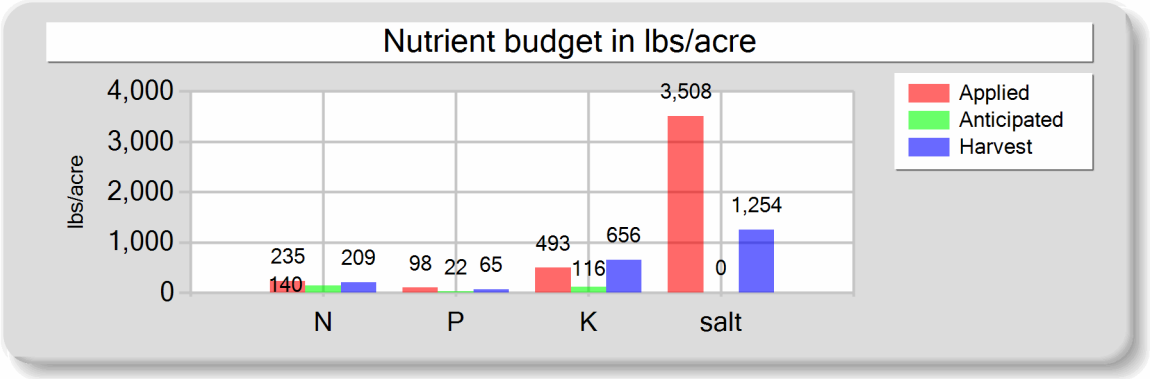
Fresh water applied	
66,246,480.00	*gallons*
2,439.63	*acre-inches*
27.11	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pivot 4 - 10/30/2022: Oats, silage-soft dough

Field name: Pivot 4 Crop: Oats, silage-soft dough Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	228.01	98.41	493.05	3,507.84
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.01	98.41	493.05	3,507.84
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	209.45	65.28	656.25	1,254.01
Nutrient balance	25.55	33.13	-163.20	2,253.83
Applied to removed ratio	1.12	1.51	0.75	2.80

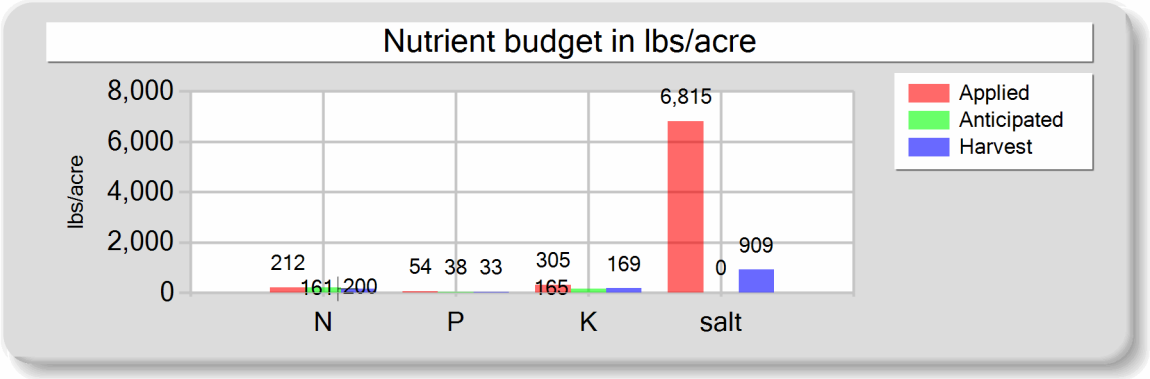
Fresh water applied	
5,532,850.00	*gallons*
203.76	*acre-inches*
4.08	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pivot 4 - 06/02/2023: Corn, silage

Field name: Pivot 4 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.80	0.00	0.00	0.00
Dry manure	166.32	54.22	304.92	6,497.57
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	6.16	0.00	0.00	317.18
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	212.28	54.22	304.92	6,814.75
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	160.65	33.24	168.96	909.43
Nutrient balance	51.63	20.98	135.96	5,905.32
Applied to removed ratio	1.32	1.63	1.80	7.49

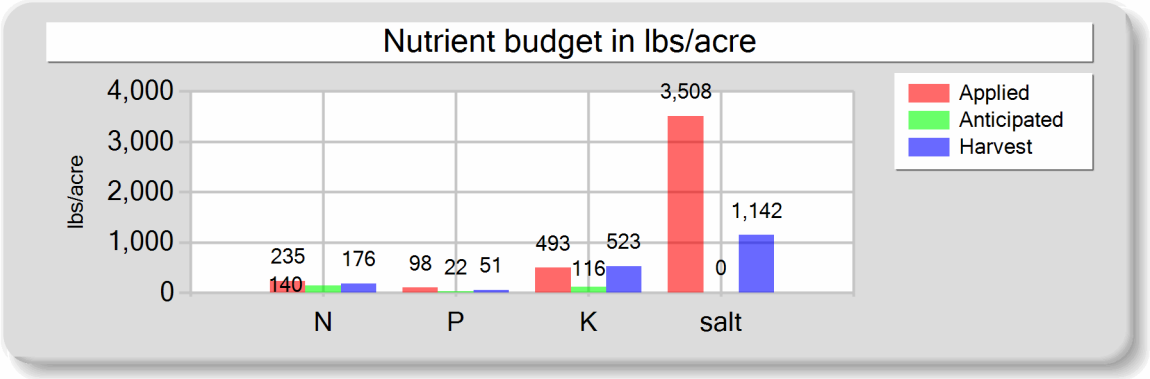
Fresh water applied	
36,546,700.00	*gallons*
1,345.89	*acre-inches*
26.92	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pivot 5 - 10/31/2022: Oats, silage-soft dough

Field name: Pivot 5 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	228.01	98.41	493.05	3,507.84
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.01	98.41	493.05	3,507.84
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	175.66	51.11	522.86	1,142.08
Nutrient balance	59.35	47.30	-29.81	2,365.76
Applied to removed ratio	1.34	1.93	0.94	3.07

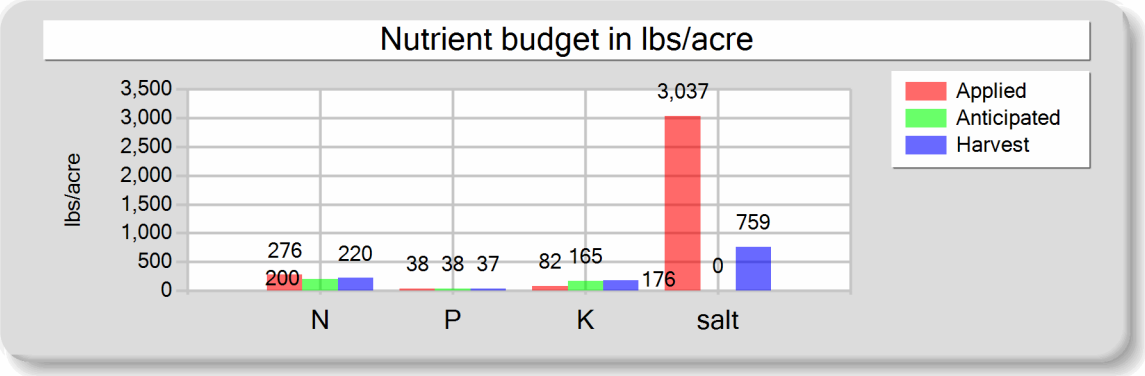
Fresh water applied	
7,746,410.00	*gallons*
285.27	*acre-inches*
4.08	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pivot 5 - 06/05/2023: Corn, silage

Field name: Pivot 5 Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.80	0.00	0.00	0.00
Dry manure	230.03	38.43	81.88	2,720.14
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	6.15	0.00	0.00	316.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	275.98	38.43	81.88	3,037.00
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	220.31	36.57	175.71	759.21
Nutrient balance	55.67	1.86	-93.83	2,277.78
Applied to removed ratio	1.25	1.05	0.47	4.00

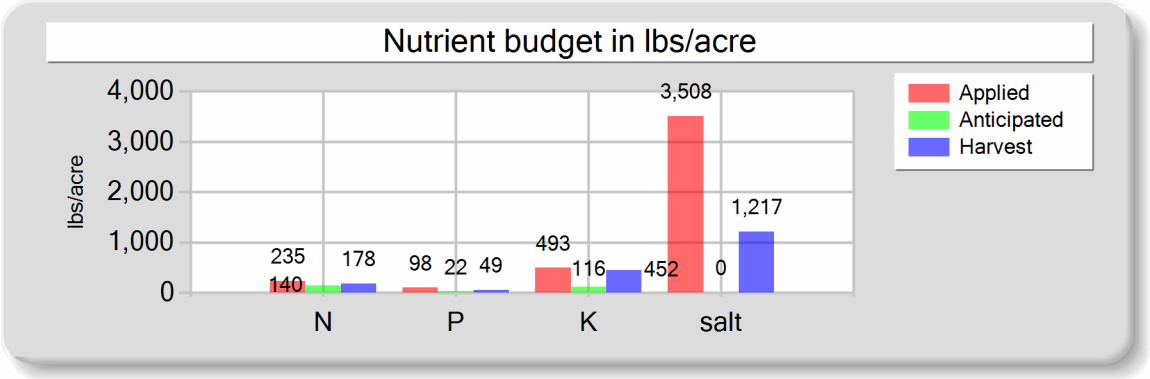
Fresh water applied	
51,112,180.00	*gallons*
1,882.29	*acre-inches*
26.89	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pivot 6 - 10/31/2022: Oats, silage-soft dough

Field name: Pivot 6 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	228.01	98.41	493.05	3,507.84
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.01	98.41	493.05	3,507.84
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	177.85	49.31	451.78	1,217.16
Nutrient balance	57.16	49.10	41.26	2,290.68
Applied to removed ratio	1.32	2.00	1.09	2.88

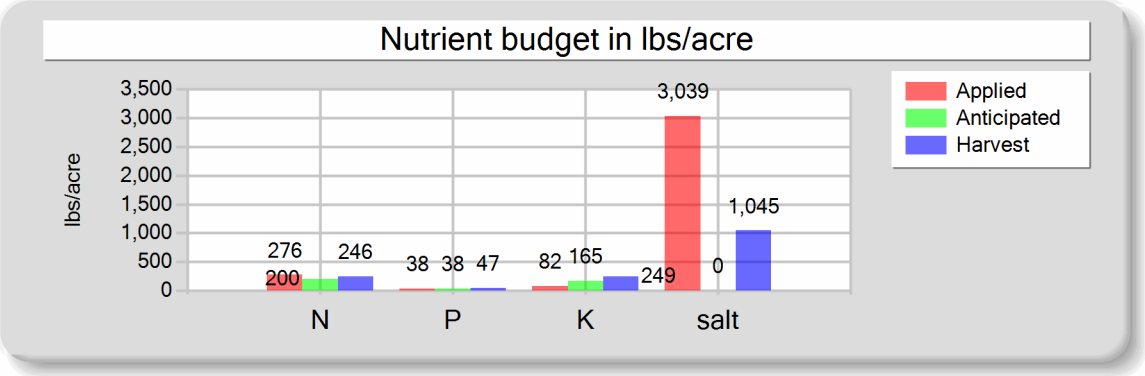
Fresh water applied	
5,527,950.00	*gallons*
203.58	*acre-inches*
4.07	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pivot 6 - 06/06/2023: Corn, silage

Field name: Pivot 6 Crop: Corn, silage Plant date: 06/06/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.80	0.00	0.00	0.00
Dry manure	230.03	38.43	81.88	2,720.14
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	6.19	0.00	0.00	318.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	276.01	38.43	81.88	3,038.67
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	246.16	46.79	249.21	1,044.87
Nutrient balance	29.85	-8.36	-167.34	1,993.80
Applied to removed ratio	1.12	0.82	0.33	2.91

Fresh water applied	
36,701,300.00	*gallons*
1,351.58	*acre-inches*
27.03	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,400.00	10,100.00	50,600.00	26,900.00	11,400.00	10,500.00	8,300.00	2,140.00		36.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Separated Solids

Sample and source description: Separated Solids

Sample date: 09/14/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 82.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,300.00	7,070.00	15,200.00							16.50
DL	100.00	100.00	100.00							0.01

Compost

Sample and source description: Compost

Sample date: 04/19/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,700.00	8,570.00	23,500.00							34.40
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,000.00	4,890.00	27,500.00	14,400.00	7,490.00	2,400.00	3,400.00	960.00		58.60
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Separated Solids

Sample and source description: Separated Solids

Sample date: 04/19/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,200.00	4,210.00	8,970.00							29.80
DL	100.00	100.00	100.00							0.01

Compost

Sample and source description: Compost

Sample date: 10/11/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 18.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,700.00	7,120.00	18,100.00							45.30
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,600.00	7,000.00	37,100.00							32.50
DL	100.00	100.00	100.00							0.01

Separated Solids

Sample and source description: Separated Solids

Sample date: 10/11/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,000.00	2,870.00	5,990.00							15.20
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ1

Sample and source description: WWQ1

Sample date: 02/23/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	616.00	384.00	0.00	0.00	86.90	953.00	170.00	135.00	227.00	3,160.00	0.00	179.00	312.00	7,130.00	6,340
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 04/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	571.00	330.00	0.00	0.00	102.00	1,090.00								6,770.00	5,920
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	529.00	370.00	0.00	0.00	43.20	919.00								6,650.00	5,060
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 11/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	837.00	283.00	0.00	0.00	118.00	560.00								5,100.00	3,380
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 11/06/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	753.00	123.00	0.00	0.00	116.00	969.00								5,630.00	2,640
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

TW Pond A

Sample and source description: TW Pond A

Sample date: 12/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	19.60	8.40	0.00	0.00	9.56	122.00								779.00	720
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

TW Pond B

Sample and source description: TW Pond B

Sample date: 12/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	47.60	19.60	0.00	0.00	21.60	480.00								2,540.00	1,820
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal OID

Canal OID

Sample description: Canal OID

Sample date: 08/18/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.01	0.00	0.00								70.90	52
DL	1.00	0.50	0.10								10.00	10

IW #1

IW #1

IW #1

Sample description: IW #1

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.40	0.00	1.30								224.00	160
DL	1.00	0.50	0.10								10.00	10

D. SOIL ANALYSES

Field 1

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			49.00				
DL			4.00				

Field 2

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			31.00				
DL			4.00				

Field 3

Field 3

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			104.00				
DL			4.00				

Field 4

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			104.00				
DL			4.00				

Field 5

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			36.00				
DL			4.00				

Field 6

Field 6

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			55.00				
DL			4.00				

Field 7

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			88.00				
DL			4.00				

Pivot 3

Soil

Sample and source description: Soil

Sample date: 05/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			41.00				
DL			4.00				

Pivot 3

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			17.00				
DL			4.00				

Pivot 4

Soil

Sample and source description: Soil

Sample date: 05/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			102.00				
DL			4.00				

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			61.00				
DL			4.00				

Pivot 5

Pivot 5

Soil

Sample and source description: Soil

Sample date: 05/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			66.00				
DL			4.00				

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			19.00				
DL			4.00				

Pivot 6

Soil

Sample and source description: Soil

Sample date: 05/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			57.00				
DL			4.00				

Pivot 6

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			30.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Field 1 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,600.00	890.00	11,700.00		10.40
DL	100.00	100.00	100.00		0.01

Field 2 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,000.00	1,060.00	9,060.00		8.75
DL	100.00	100.00	100.00		0.01

Field 3 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	1,110.00	10,800.00		9.59
DL	100.00	100.00	100.00		0.01

Field 4 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,250.00	1,130.00	9,030.00		10.00
DL	100.00	100.00	100.00		0.01

Field 5 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	1,300.00	10,700.00		11.30
DL	100.00	100.00	100.00		0.01

Field 6 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,470.00	1,090.00	9,430.00		15.00
DL	100.00	100.00	100.00		0.01

Field 7 - 11/25/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,840.00	1,020.00	6,740.00		10.20
DL	100.00	100.00	100.00		0.01

Pivot 3 - 10/30/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,030.00	1,080.00	9,980.00		8.89
DL	100.00	100.00	100.00		0.01

Pivot 3 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,600.00	730.00	3,020.00		4.60
DL	100.00	100.00	100.00		0.01

Pivot 4 - 10/30/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,160.00	1,920.00	19,300.00		9.22
DL	100.00	100.00	100.00		0.01

Pivot 4 - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,060.00	840.00	4,270.00		6.80
DL	100.00	100.00	100.00		0.01

Pivot 5 - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,980.00	1,740.00	17,800.00		9.17
DL	100.00	100.00	100.00		0.01

Pivot 5 - 06/05/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,940.00	820.00	3,940.00		4.48
DL	100.00	100.00	100.00		0.01

Pivot 6 - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,590.00	1,550.00	14,200.00		9.86
DL	100.00	100.00	100.00		0.01

Pivot 6 - 06/06/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,840.00	920.00	4,900.00		4.80
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

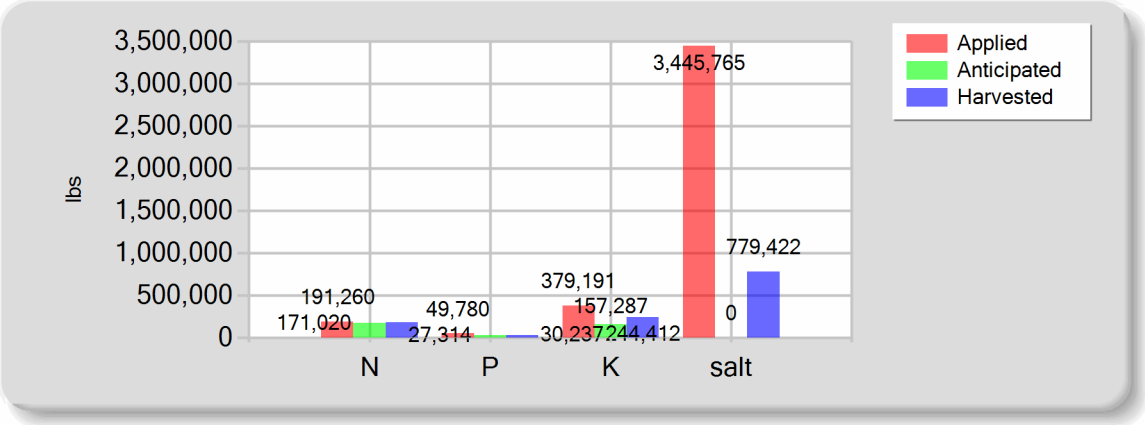
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

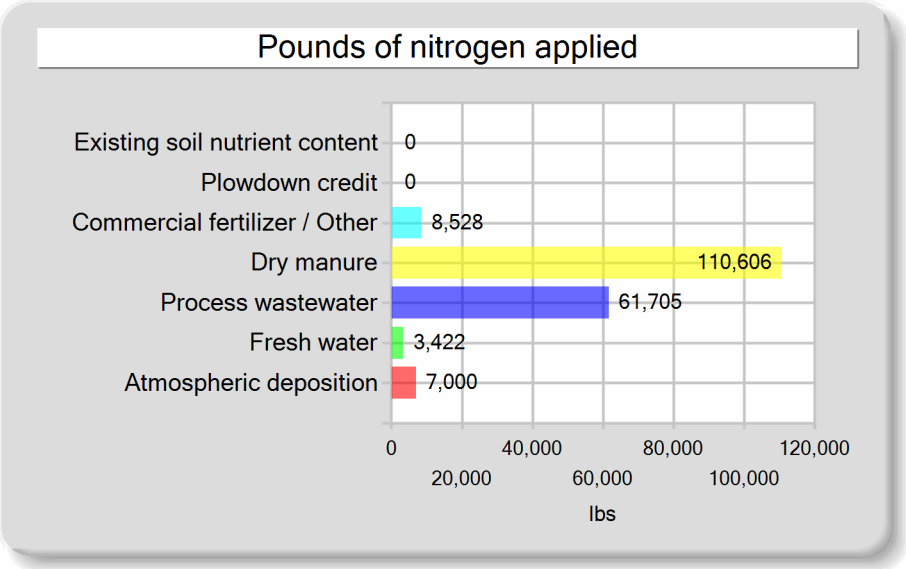
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

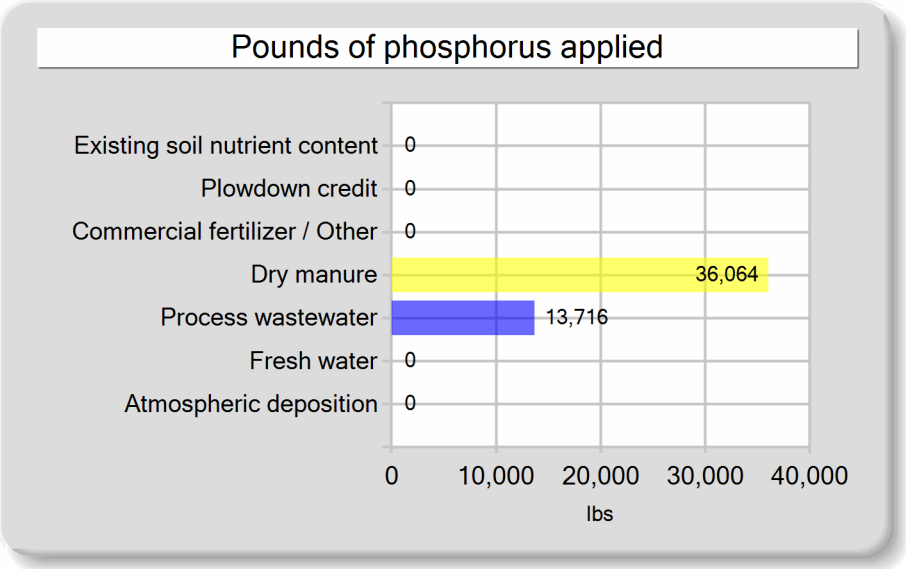
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8,528.00	0.00	0.00	0.00
Dry manure	110,606.11	36,064.17	159,102.74	2,030,848.80
Process wastewater	61,704.57	13,716.12	220,088.44	1,124,753.95
Fresh water	3,421.71	0.00	0.00	290,161.88
Atmospheric deposition	7,000.00	0.00	0.00	0.00
Total nutrients applied	191,260.38	49,780.29	379,191.18	3,445,764.63
Anticipated crop nutrient removal	171,020.00	27,314.00	157,287.00	0.00
Actual crop nutrient removal	182,378.77	30,237.00	244,411.87	779,422.12
Nutrient balance	8,881.61	19,543.29	134,779.31	2,666,342.52
Applied to removed ratio	1.05	1.65	1.55	4.42

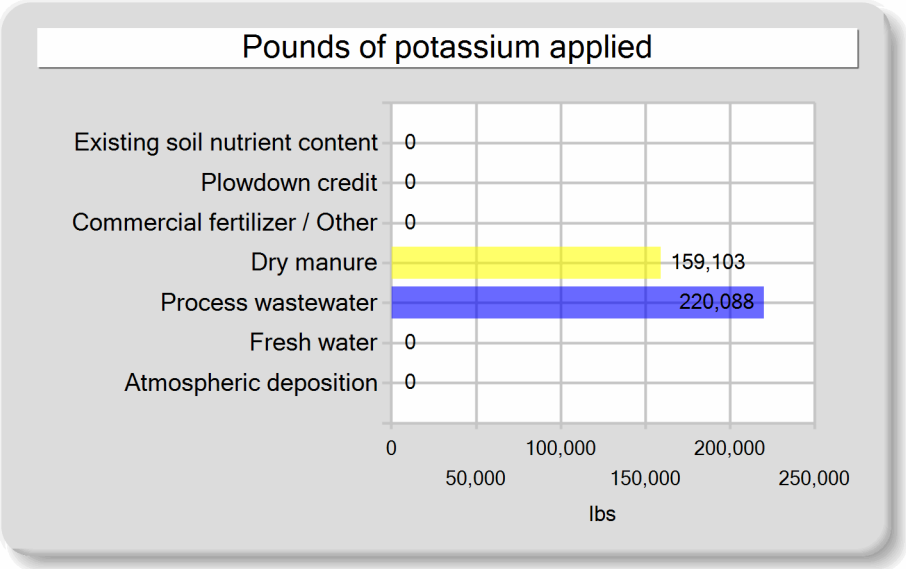
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

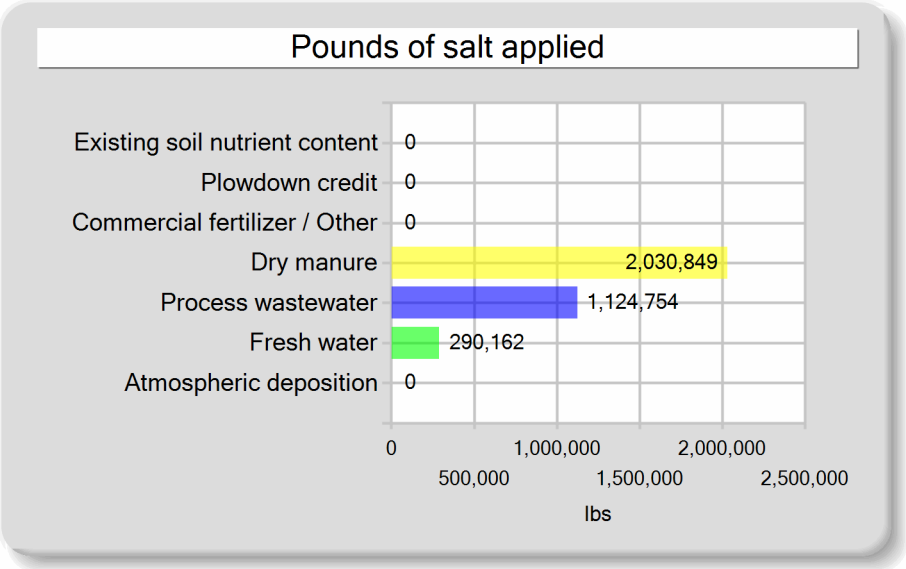


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Holly D Jongsma Holly D Jongsma (Jun 27, 2024 13:44 PDT)	Kurt Hoekstra (Jun 27, 2024 14:13 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Holly Jongsma	Kurt Hoekstra
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 27, 2024	Jun 27, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Kurt Hoekstra

Name of Dairy Facility: Cross View

Facility Address: 12530 Waznesville Rd (Number and Street) Oakdale (City) 95361 (Zip Code)

Contact Person Name and Phone Number: Kurt Hoekstra (Name) 209 765-2932 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: ALP

Address of Hauling Company /Person: 2324 Villa Manucha (Number and Street) Newman (City) 95360 (Zip Code)

Contact Person: Jeff Lima (Name) 209 862-3393 (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above): 209

Nate Ludlow (Name) 6648 Smith Rd (Number and Street) Oakdale (City) 95361 (Zip Code) 848-1844 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same as above (Number and Street) City Zip Code 010-016-013 (Assessor's Parcel Number)

Dates Hauled: 7/28/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 1540 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 18.4 % Moisture

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: 154 Loads @ 10 T/Load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Kurt Hoekstra (Jun 27, 2024 14:13 PDT) Date: Jun 27, 2024

Hauler's Signature: _____ Date: 7-31-23

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Kurt Hoekstra

Name of Dairy Facility: Cross View

Facility Address: 12530 Waznerville Rd (Number and Street) Oakdale (City) 95361 (Zip Code)

Contact Person Name and Phone Number: Kurt Hoekstra (Name) 209 765-2932 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: ALP

Address of Hauling Company /Person: 2324 Villa Manucha (Number and Street) Newman (City) 95360 (Zip Code)

Contact Person: Jeff Lima (Name) 209 862-3393 (Phone Number)

Destination Information:

Composting Facility / Broker / (Farmer) / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above): 209

AL Gilbert V Ranch (Name) 12623 Fogarty Rd (Number and Street) Oakdale (City) 95361 (Zip Code) 847-1721 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same as above (Number and Street) _____ (City) _____ (Zip Code) 010-072-001 (Assessor's Parcel Number)

Dates Hauled: 9/12/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 3200 (Tons) or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 18.4 % Moisture

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

320 Loads @ 10 T/Load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Kurt Hoekstra (Jun 27, 2024 14:13 PDT) Date: Jun 27, 2024

Hauler's Signature: _____ Date: 9-15-23

12/14/23 14:20 23L0958

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6120 • 800 322-0996 • Fax 559 233-8136

Purchase Order No | Bill To: Acct # 26363 | Cons #

No. Samples: 2 | No of Bottles: 3

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater [ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: Pear View
Address: 12530 Wasserville Rd
City: Oakdale State: CA Zip: 95361
Telephone: 209 847-5877 Fax:
Cell/Email:
COPY TO: F&H Ag Services
E.Sawyer@Fnaservices.com
REQUESTED BY: Mitch Sawyer
PROJECT:
CROP:

Other:

Analysis and Bottles Required: (Please Indicate Analysis)

DCWI + TU
DWWI

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Filed ID
1	IW 1	12/13/23	Started 6:30 AM	11.3	
2	DW Barn / House	↓		11.4	
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221415957
Correction Factor: 0°C
Calibration Date: 02/06/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Mitch Sawyer	F&H Ag Services		
Second	[illegible]	[illegible]	12/14/23 [illegible]	12/14/23 2:15
Third				
Fourth	[illegible]	[illegible]	12/14 16:20	

I guarantee that as the client, or on behalf of client, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be cause for suit to be brought, reasonable attorneys' fees. It is understood that payment is expected to be made within 30 days from the date of invoice unless other terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $3.00 per month, whichever is greater.
If payment is not made when due and a legitimate dispute arises concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Dispute Alternative to Litigation, Inc. (ADL). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ADL under its Rules and Procedures. The parties will equally share the cost of the arbitration. If, however, any party fails to pay its share of the arbitration cost, that party will pay the entire cost of the arbitration and, as to that party, waive the right to be represented by reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature
Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

pH Strips
Lot 16PDH0501
Exp. Jan. 2025

DELLAVALLE
LABORATORY INC

Report of Dairy Canal Water Analysis

Cross View
12530 Warnerville Rd
Oakdale CA 95361
00-0026363 13

E-mail: jsawyer@fragservices.com
Copy To:

Lab No.: 23L0958
Sampled By:
Requested By: Jeanette Sawyer
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	TDS mg/L	TKN mg/L	Total Nitrogen TN mg/L
1	IW 1	12/13/23	6:30	224	0.22	1.3	160	ND	1.40

ND = None Detected

Approved By: Brent McFriedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE
LABORATORY INC

Report of Dairy Well Water Analysis

Lab No.: 23L0958
Sampled By:
Requested By: Jeanette Sawyer
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

Cross View
12530 Warnerville Rd
Oakdale CA 95361
00-0026363 13

E-mail: jsawyer@fragservices.com
Copy To:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
2	DW Barn/House	12/13/23	6:30	250	0.25	1.8	*		8.1

* = Field NH_4-N not Taken.
ND = None Detected

Approved By: Erica M Grieilland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

2023 Annual Report - Cross View Dairy

Final Audit Report 2024-06-27

Created:	2024-06-27
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAA-GDiPsvFYg3lV0Fgy3FR1TSoj5oEA9dC

"2023 Annual Report - Cross View Dairy" History

Document created by Vince Furtado (vfurtado@fragservices.com)
2024-06-27 - 7:01:55 PM GMT

Document emailed to Holly D. Jongsma (wranglerjranch@comcast.net) for signature
2024-06-27 - 7:04:18 PM GMT

Email viewed by Holly D. Jongsma (wranglerjranch@comcast.net)
2024-06-27 - 8:34:58 PM GMT

Signer Holly D. Jongsma (wranglerjranch@comcast.net) entered name at signing as Holly D Jongsma
2024-06-27 - 8:44:08 PM GMT

Document e-signed by Holly D Jongsma (wranglerjranch@comcast.net)
Signature Date: 2024-06-27 - 8:44:10 PM GMT - Time Source: server

Document emailed to hoekstrakurt@gmail.com for signature
2024-06-27 - 8:44:12 PM GMT

Email viewed by hoekstrakurt@gmail.com
2024-06-27 - 9:12:09 PM GMT

Signer hoekstrakurt@gmail.com entered name at signing as Kurt Hoekstra
2024-06-27 - 9:13:34 PM GMT

Document e-signed by Kurt Hoekstra (hoekstrakurt@gmail.com)
Signature Date: 2024-06-27 - 9:13:36 PM GMT - Time Source: server

Agreement completed.
2024-06-27 - 9:13:36 PM GMT